Final CS 6210

You may talk about problems with the TA and with me, or reference any relevant literature (including the Julia documentation!), providing attribution for any good ideas you might get. You should not consult with other people or with AI tools, and your final write-up should be your own.

```
1 using LinearAlgebra
```

1. Truth or dare

State whether each of the following is true and give a short argument why or why not. (One point each)

- In the Tikhonov-regularized least squares problem, choosing the regularization parameter λ via leave-one-out cross validation minimizes the overall residual norm.
- If $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite and $A = R^T R$ is the Cholesky factorization, then the column two-norms of R satisfy $\lambda_n(A) \leq \|R_{:,j}\|^2 \leq \lambda_1(A)$, where $\lambda_1(A)$ and $\lambda_n(A)$ are the largest and smallest eigenvalues of A, respectively.
- Suppose $A \in \mathbb{R}^{n \times n}$ is written as $A = M - N$ where M is invertible. If the iteration $Mx^{(k+1)} = Nx^{(k)} + b$ converges to a limiting fixed point x^*, then $Ax^* = b$.
- If A is diagonalizable and p is a polynomial such that $p(\lambda) = \lambda^{-1}$ for every eigenvalue λ in the spectrum of A, then $p(A) = A^{-1}$.

1. False. Leave-one-out cross-validation (LOOCV) selects λ to minimize the average prediction error on held-out data points, balancing bias and variance for better generalization (as in the bias-variance decomposition). It does not minimize the training residual $\|Ax - b\|^2$, which is smallest at $\lambda = 0$.
2. True. For SPD A, the Rayleigh quotient gives $\lambda_1(A) = \max_{\|x\|=1} x^T A x \geq e_j^T A e_j = A_{jj}$ for all j (so $A_{jj} \leq \lambda_1(A)$), and $\lambda_n(A) = \min_{\|x\|=1} x^T A x \leq A_{jj}$ (so $A_{jj} \geq \lambda_n(A)$). From Cholesky, $\|R_{:,j}\|^2 = e_j^T R^T R e_j = e_j^T A e_j = A_{jj}$. Thus, the bounds hold for all j.
3. True. At the fixed point, $Mx^* = Nx^* + b$, so $(M - N)x^* = b$, i.e., $Ax^* = b$. This holds regardless of convergence rate, as long as a limit exists.
4. True. Since $A = V \Lambda V^{-1}$ (diagonalizable), $p(A) = V p(\Lambda) V^{-1}$ where $p(\Lambda) = diag(p(\lambda_i)) = diag(\lambda_i^{-1}) = \Lambda^{-1}$. Thus, $p(A) = V \Lambda^{-1} V^{-1} = A^{-1}$.

2. Scaled iteration

Consider the stationary iteration

$$x^{(k+1)} = x^{(k)} + M^{-1}A^T(b - Ax^{(k)})$$

where $A \in \mathbb{R}^{m \times n}$ has full column rank and $M \in \mathbb{R}^{n \times n}$ is positive definite.

- (1 point): What is the fixed point?
- (1 point): Write an iteration equation for the error
- (2 points): Argue that if $\|Ax\|^2 / \|x\|_M^2 < 2$ for all $x \neq 0$, then the iteration converges.

1. Fixed point: Set $x^* = x^{(k+1)} = x^{(k)}$, so $0 = M^{-1}A^T(b - Ax^*)$. Since M is invertible, $A^T(b - Ax^*) = 0$, the normal equations for least squares. Thus, $x^* = (A^TA)^{-1}A^Tb$ (exists since A full column rank). The fixed point is independent of M.
2. Iteration for error: Let $e^{(k)} = x^{(k)} - x^*$. Then $e^{(k+1)} = e^{(k)} + M^{-1}A^T(b - Ax^{(k)}) = e^{(k)} + M^{-1}A^T((b - Ax^*) - Ae^{(k)})$. Since $A^T(b - Ax^*) = 0$, this simplifies to $e^{(k+1)} = e^{(k)} - M^{-1}A^TAe^{(k)} = (I - M^{-1}A^TA)e^{(k)}$.
3. Convergence if $\|Ax\|^2 / \|x\|_M^2 < 2$ for all $x \neq 0$: The iteration matrix is $G = I - M^{-1}A^TA$, and convergence requires $\rho(G) < 1$. The eigenvalues of $M^{-1}A^TA$ are μ where $\mu = x^TA^TAx / x^TMx = \|Ax\|^2 / \|x\|_M^2$ (Rayleigh quotients). Since A full column rank and $M > 0$, $\inf \mu > 0$; given $\sup \mu < 2$, all $\mu \in (0, 2)$. Thus, eigenvalues of G are $1 - \mu \in (-1, 1)$, so $\rho(G) < 1$. Let $B = AM^{-1/2}$; the condition is $\sigma_{\max}(B)^2 < 2$, confirming clustered eigenvalues for fast convergence.

3. Factorization free

Given an initial guess $x^{(0)}$ with $|x^{(0)} - \lambda^{-1}| < |\lambda^{-1}|$ the Newton iteration

$$x^{(k+1)} = x^{(k)} + x^{(k)}(1 - \lambda x^{(k)})$$

converges quadratically to λ^{-1}; more specifically, the error $e^{(k)} = x^{(k)} - \lambda^{-1}$ satisfies

$$e^{(k+1)} = -\lambda(e^{(k)})^2.$$

Similarly, the matrix iteration

$$X^{(k+1)} = X^{(k)} + X^{(k)}(I - AX^{(k)})$$

converges quadratically to A^{-1} given a good initial guess.

(1 point): Argue that for the initial guess $x^{(0)} = 2 - \lambda$, the scalar Newton iteration converges for $|1 - \lambda| < 1$. You may take the convergence condition given at the start of the problem as true without proof.

(1 point): Argue that if A is diagonalizable and $X^{(0)} = p_0(A)$ for some polynomial p_0, then the matrix Newton iteration converges iff the scalar Newton iteration converges for each eigenvalue λ of A starting from $p_0(\lambda)$.

(2 points): Argue that if $\|I - A\| < 1$ for some operator norm, then the matrix Newton iteration converges starting from the initial guess $X^{(0)} = 2I - A$

Scalar Newton: $x^{(k+1)} = x^{(k)} + x^{(k)}(1 - \lambda x^{(k)}) = x^{(k)}(2 - \lambda x^{(k)})$, converges quadratically if $|x^{(0)} - \lambda^{-1}| < |\lambda^{-1}|$, with $e^{(k+1)} = -\lambda(e^{(k)})^2$.

1. For $x^{(0)} = 2 - \lambda$, converges if $|1 - \lambda| < 1$ (i.e., $\lambda \in (0, 2)$): Compute $e^{(0)} = (2 - \lambda) - \lambda^{-1} = 2 - \lambda - \lambda^{-1}$. Need $|2 - \lambda - \lambda^{-1}| < \lambda^{-1}$ (since $\lambda > 0$). Let $f(\lambda) = 2 - \lambda - \lambda^{-1} \leq 0$ for $\lambda \in (0, 2)$, so $|f| = \lambda + \lambda^{-1} - 2$. Then $|f| - \lambda^{-1} = \lambda - 2 < 0$ (strict since $\lambda < 2$). Thus, the condition holds.
2. If A diagonalizable and $X^{(0)} = p_0(A)$, matrix Newton converges iff scalar does for each λ_i from $p_0(\lambda_i)$: Let $A = V\Lambda V^{-1}$; then $X^{(0)} = Vp_0(\Lambda)V^{-1}$. The recursion $X^{(k+1)} = X^{(k)}(2I - AX^{(k)})$ preserves diagonalization: by induction, $X^{(k)} = VY^{(k)}V^{-1}$ where $Y_i^{(k+1)} = Y_i^{(k)}(2 - \lambda_i Y_i^{(k)})$, the scalar recursion. Convergence (quadratic) holds iff each scalar does.
3. If $\|I - A\| < 1$ for some operator norm, converges from $X^{(0)} = 2I - A$: The norm bound implies $\rho(I - A) < 1$, so $\max_i |1 - \lambda_i| < 1$ (all $\lambda_i \in (0, 2)$). By (ii), since diagonalizable not assumed but $X^{(0)}$ commutes with A (preserves simultaneous diagonalizability if needed; convergence follows from spectral radius). Each scalar starts at $2 - \lambda_i$.

4. Rational extrema

Consider a rational function of the form

$$f(z) = c^T(zI - A)^{-1}b + d$$

Note that this can also be written via a Schur complement in the matrix

$$\begin{bmatrix} A - zI & b \\ c^T & d \end{bmatrix} = M - zD$$

The poles of f are the eigenvalues of A and the zeros are the eigenvalues of $M - zD$ (as discussed in HW 10).

(1 point): Argue that if $d \neq 0$, then the zeros of f are the eigenvalues of $A - bc^T/d$.

(1 point): Write a function to evaluate the function and derivative in $O(n^2)$ time given a Hessenberg factorization object `F = hessenberg(A)`. Your code should take $O(n^2)$ time, and should look like

```
f, df = p4eval(z, F, b, c, d)
```

(1 point): Introducing variables $v = (zI - A)^{-1}b$ and $u = (zI - A)^{-1}v$, write a generalized eigenvalue problem whose solutions correspond to critical points of $f(z)$.

(1 point): Using `eigvals(A,B)` (which computes solutions to the generalized eigenvalue problem $Ax = \lambda Bx$), find the (real) critical points of f. Your code should look like

```
zs = p4critical(A, b, c, d)
```

Note that you should write testers for each code.

```
p4eval (generic function with 1 method)
```

```
function p4eval(z, F, b, c, d)
    n = size(F.H, 1)
    Id = I(n)
    Q = F.Q
    H = F.H
    qb = Q' * b
    w = (z * Id - H) \ qb
    v = Q * w
    f = c' * v + d
    uu = (z * Id - H) \ w
    df = -c' * (Q * uu)

    f, df
end
```

```
begin
    n = 5; A = randn(n,n); F = hessenberg(A); b = randn(n); c = randn(n); d = 1.0; z = 10.0
    f, df = p4eval(z, F, b, c, d)
    f_direct = c' * ((z*I - A) \ b) + d
    df_direct = -c' * ((z*I - A) \ ((z*I - A) \ b))
    print(abs(f - f_direct) < 1e-10 && abs(df - df_direct) < 1e-10)
end
```

```
true
```

1. If $d \neq 0$, zeros are eigenvalues of $A - bc^T/d$: Using the matrix determinant lemma https://en.wikipedia.org/wiki/Matrix*determinant*lemma, $\det(zI - A + bc^T/d) = \det(zI - A)(1 + (c^T/d)(zI - A)^{-1}b) = \det(zI - A)f(z)/d$. Zeros of $f(z)$ (non-poles) are eigenvalues of $A - bc^T/d$.
2. Function for f, f' using Hessenberg $F = hessenberg(A)$. $A = QHQ^T$, so $(zI - A)^{-1}b = Q(zI - H)^{-1}(Q^Tb)$; $f' = -c^T(zI - A)^{-2}b = -c^T(zI - A)^{-1}v$. Each is $O(n^2)$ for Hessenberg (LU factorization efficient).

```
p4critical (generic function with 1 method)
```

```
function p4critical(A, b, c, d)
    n = size(A, 1)
    S = zeros(2n + 1, 2n + 1)
    # Order: v (1:n), u (n+1:2n), s (2n+1)
    S[1:n, 1:n] = A # A v
    S[1:n, 2n+1] = b # b s
    S[n+1:2n, n+1:2n] = A # A u
    S[n+1:2n, 1:n] = I(n) # v
    S[2n+1, n+1:2n] = c' # c^T u
    L = zeros(2n + 1, 2n + 1)
    L[1:n, 1:n] = I(n)
    L[n+1:2n, n+1:2n] = I(n)
    evs = eigvals(S, L)
    finite_real = filter(e -> isfinite(e) && isreal(e), evs)
    zs = sort(real.(finite_real))

    zs
end
```

```
begin
    n2 = 3; A2 = diagm([1.0, 2.0, 3.0]); b2 = ones(n2); c2 = [1, -1, 1]; d2 = 1.0
    zs = p4critical(A2, b2, c2, d2)
    F2 = hessenberg(A2)
    for z in zs
        _, df = p4eval(z, F2, b2, c2, d2)
        print(abs(df) < 1e-10)
    end
end
```

```
truetrue
```

3. Critical points satisfy $f'(z) = 0$, i.e., $-c^T(zI - A)^{-2}b = 0$. Let $v = (zI - A)^{-1}b$, so $(zI - A)v = b$. Let $u = (zI - A)^{-1}v$, so $(zI - A)u = v$ and $c^T u = 0$. Introduce scalar s for homogeneity: the GEP is $\begin{bmatrix} A & 0 & b \\ I & A & 0 \\ 0 & c^T & 0 \end{bmatrix} \begin{bmatrix} v \\ u \\ s \end{bmatrix} = z \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ u \\ s \end{bmatrix}$ Finite eigenvalues z where $s \neq 0$ are critical points (normalize $s = 1$).
4. eigvals(S, L) solves the GEP. Filters finite real eigenvalues (infinite ones from singular L) and sorts them.

5. Playing with polynomials

Suppose $A = I - F$ where $\|F\|_2 < 1$.

(2 points): Find $p \in \mathcal{P}_d$ such that $\|I - Ap(A)\|_2 \leq \|F\|_2^{d+1}$.

(2 points): Given a particular F, write a function `polyinv(F,d)` to compute the $d + 1$ coefficients with respect to the power basis for the polynomial $p \in \mathcal{P}_d$ such that $\|I - Ap(A)\|_F^2$ is minimal. Also

give an explanation for your code!

```
polyinv (generic function with 1 method)
```

```
function polyinv(F, d)
    n = size(F, 1)
    A = I(n) - F
    Phi = zeros(n*n, d+1)
    A_power = A
    for i in 1:(d+1)
        Phi[:, i] = vec(A_power)
        A_power = A_power * A
    end
    target = vec(Matrix(I(n)))
    c = Phi \ target

    c
end
```

1. Consider the polynomial $p(x) = \sum_{k=0}^{d}(1-x)^k \in \mathcal{P}_d$. Since $A = I - F$, we have $p(A) = \sum_{k=0}^{d} F^k$ Then, $Ap(A) = (I-F)\sum_{k=0}^{d} F^k = \sum_{k=0}^{d} F^k - \sum_{k=1}^{d+1} F^k = I - F^{d+1}$, so $I - Ap(A) = F^{d+1}$. Taking the 2-norm yields $\|I - Ap(A)\|_2 = \|F^{d+1}\|_2 \leq \|F\|_2^{d+1}$ as required.

2. The minimization is reformulated as a standard linear least-squares problem by vectorizing the matrices. Since the Frobenius norm satisfies $\|M\|_F = \|\mathbf{vec}(M)\|_2$, we have $\|I - Ap(A)\|_F = \left\|I - \sum_{k=0}^{d} c_k A^{k+1}\right\|_F = \left\|\sum_{k=0}^{d} c_k A^{k+1} - I\right\|_F = \|\Phi\mathbf{c} - \mathbf{vec}(I)\|_2$ where Φ is the $n^2 \times (d+1)$ matrix whose columns are $\mathbf{vec}(A), \mathbf{vec}(A^2), \ldots, \mathbf{vec}(A^{d+1})$, and $\mathbf{c} = [c_0; c_1; \ldots; c_d]$. The code solves for the $\mathbf{c}$ that minimizes this via QR factorization. Phi = zeros(n*n, d+1): Allocates the design matrix Φ as a dense $n^2 \times (d+1)$ array of zeros. A*power = A: Initializes a temporary matrix to hold powers of A, starting with $A^1 = A$. Phi[:, i] = vec(*A*power): Flattens the current power A^i into a column vector and assigns it to the i-th column of Φ. target = vec(Matrix(I(n))): Vectorizes the identity matrix into an $n^2 \times 1$ target vector. c = Phi \ target: Solves the least-squares system $\Phi\mathbf{c} \approx \mathbf{vec}(I)$.